

Week 2 – Design Studio: Answers

"A résumé screening system rejected a qualified candidate. Write the requirements it should have had, and design the measurement that would have told you."

Read this after the studio. It isn't the only good answer; it's what a strong answer contains, in the order the pieces depend on each other.

Two answers that are not answers

"Make the model more accurate" has no content until you say: accurate on whom, at what rate, measured on what, and who agreed to that number. A requirements table with no plan to measure it is the same thing in a different form: every line in it is a wish. A complete answer has both halves on the page.

What a strong answer contains

1. Frame the decision before you write a rate

What does the model decide? If it *rejects*, then a wrong output is an irreversible harm to a person who never learns it happened. If it *ranks and routes* — advance, human review, hold — then the worst it can do alone is put a qualified candidate in a review pile. You can only write sensible rates for the second system. Say which one you are specifying.

2. The requirements, in the language of rates

RATE	SLICE	REMAINDER POLICY	OWNER
False-reject rate $\leq 5\%$	each slice, including ones the model cannot see directly: career gap over 12 months, non-traditional education, non-native phrasing	every low-ranked candidate on a monitored slice is human-reviewed before any rejection	Head of talent acquisition
Advance rate $\geq 80\%$	candidates a panel labelled qualified	misses are reviewed and added to the golden set	Head of talent acquisition
Score changes by ≤ 2 points when only a name, a college name, or the word "women's" changes	matched pairs of résumés	any pair that moves more is an incident, not a statistic	Legal / compliance
Reject without a human = 0	all candidates	any occurrence is an incident	Legal / compliance

Two things to notice. The last row is a rate of **zero**: no sample can show it, so it cannot be a property of the model. It is a property of the code around the model — the reject action does not exist without a human's click. And the third row is a different *kind* of requirement: it is about pairs, not about single résumés, and it is the one that would have caught Amazon's system.

3. Where the right answers come from — the hard part

You need résumés with a known right answer. The obvious source is wrong:

- **"People we hired who did well"** tells you nothing about people you rejected. The labels exist only where the old process said yes. A model evaluated on them can look excellent while repeating every past mistake. This is a selection problem: your own outputs chose which labels exist.
- **Better**: a panel labels a random sample of *all* applications, including ones the current process rejected, blind to the model's score. Expensive. It is the cost of knowing.
- **Who decides "qualified"**? Two panelists will disagree. Measure how often. A résumé they cannot agree on is tagged ambiguous and kept; it is telling you the requirement is not yet well defined.

4. How many, per slice — do the arithmetic

This half is easy to leave out, and tonight's lecture is the tool for it.

- You require false-reject $\leq 5\%$ on "career gap over 12 months." Your golden set has 30 qualified résumés on that slice, and the model wrongly rejects none. **Zero in 30 bounds the rate at about 10%** (the rule of three: $3/30$). You have not shown 5%. You need about 60 clean résumés to bound it at 5%,

and if you expect a failure or two, more: 2 wrong in 100 gives a 95% interval of about 0.6% to 7%, which still straddles your requirement.

- The slices that matter most are the rarest, so a random sample will starve them. **Oversample the slices** when you build the golden set, and report each slice separately. Never pool them back into one number.
- Report counts. "28 of 30" carries its own warning. "93%" does not.

5. The measurement, one row per requirement

The requirements table is a list of wishes until each row has a measurement bolted to it. Same rows, same order. For each requirement: where the right answers come from, how many you need before the number means anything, and what you actually count.

REQUIREMENT	WHERE THE RIGHT ANSWERS COME FROM	HOW MANY	WHAT YOU RUN AND COUNT
False-reject $\leq 5\%$ on each monitored slice	The panel-labelled golden set: a random sample of all applications, labelled blind to the model. The panel also tags each résumé's slices, because the model cannot see them and neither can a filter. Oversample every monitored slice.	60 panel-qualified résumés per slice with no misses bounds the rate at 5%. Plan for about 100, since you will see a miss or two. 30 is not enough.	One run of the screener over the golden set, outputs saved. Scorer: a panel-qualified résumé routed to <i>hold</i> is a miss. Misses over qualified, per slice .
Advance $\geq 80\%$ on panel-qualified candidates	The same golden set, qualified subset.	A few hundred. At 250 qualified: 220 advanced is 88% (83%–92%), met. 210 advanced is 84% (79%–88%), cannot tell, because the interval still covers 80%.	The same saved run. Scorer: a panel-qualified résumé routed to <i>advance</i> is a hit. Hits over qualified, overall and per slice.
Score moves ≤ 2 points when only a name, a college name, or the word "women's" changes	No panel needed. Copy real résumés from the golden set and change exactly one thing: a name coded female or male, one college for another, "women's" added to or removed from a club or award. The right answer is known by construction: no change.	50 pairs per perturbation type , about 150 in all. This is a zero-tolerance requirement, so the count is for detection, not a rate. 50 is also enough to see a consistent 1-point shift that no single pair would show.	Both members of every pair in the same run, same model version. Needs the score, not only the route. Scorer: the absolute difference between the two scores. Count pairs over 2 points, and record the mean signed shift per type.
Reject without a human = 0	No golden set. This is a rule about the code, not the model: no rejection record exists without a reviewer's identity on it.	All of them. A census, not a sample. No number of samples proves a zero.	A unit test on every build: the reject path fails without a reviewer id. A query over the production audit log: rejections with no reviewer id.

The report is one line per row above, and one per slice where the row has slices. Counts first, then the rate with its interval, then the requirement, then the verdict:

REQUIREMENT	SLICE	COUNT	RATE (95% INTERVAL)	REQ	VERDICT
False-reject	career gap	2 held of 74	2.7% (0.7%–9.3%)	≤ 5%	cannot tell
False-reject	non-traditional education	0 held of 88	0.0% (0.0%–4.2%)	≤ 5%	met
Advance	all qualified	220 of 250	88% (83%–92%)	≥ 80%	met
Pairs	female name	3 of 50 over 2 pts	mean shift –1.4	0	incident
Pairs	college name	0 of 50 over 2 pts	mean shift +0.1	0	met
Reject w/o human	all	0 of 1,812	—	0	met

Two things to notice. The first two rows are **one run, two scorers**: the same golden set and the same saved outputs, counted two ways. The third row is the only one that would have caught Amazon's system, and it is also the cheapest to build, because each pair is its own control. Comparing a résumé to a copy of itself is far more sensitive than comparing two separate groups of résumés.

Mechanics that apply to every row:

- **Record once, score many times.** Run the screener over the golden set and save every output. The scorers read the saved outputs.
- **The noise floor.** If the screener is a deterministic model, re-running it changes nothing: the noise floor is zero. That does **not** make the number certain. All the uncertainty is in which résumés you happened to label, and only more résumés shrinks it. If the screener calls a language model, run the suite five times and report the spread per slice before you compare anything to anything.
- **The report.** One row per slice: passed, total, the interval, the requirement, and met / not met / **cannot tell**. "Cannot tell" is a legitimate result. It is a purchase order for more labels.

6. What this measurement cannot see

- Slices nobody thought to cut.
- Candidates who never applied because of how the posting was worded.
- Whether the panel's idea of "qualified" is itself biased: the harness measures agreement with the panel, not with the truth.
- Drift: the golden set describes last year's applicants.

Write these down. A short honest list is worth more than a confident number.

Check your sketch

Did you say what the model decides? Did every rate have a slice and a name? Was one of them zero, and did you say where it is enforced? Did you say where labelled résumés come from, and why "people we hired" is the wrong answer? Did every requirement have its own row: where its right answers come from, how many, what you count, what the report line says? Did you put a number on how many résumés a slice needs? Did your report have a "cannot tell" column? Did you name one thing the measurement cannot see?